

Valuation & Multiples

Practice Problem Solutions

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These three problems will help you practice valuation using both the discounted cash flow methodology and multiples.

1. Swatch, Inc. is thinking of acquiring Timex, Inc., and Swatch is attempting to determine the highest price per share it should be willing to pay for Timex. It expects Timex's FCF will equal \$100 million next year (i.e., $t = 1$), \$150 million the year after (i.e., $t = 2$), and then grow at 10 percent a year thereafter and continue doing so forever. While Timex has \$10 million in cash on hand now, all this cash is required for operations of the firm. Timex also has \$900 million in debt and 100 million outstanding shares. Timex's WACC is 15 percent.
 - a. Using the constant growth assumption given above, what is a terminal value of Timex in year 2 that you can use in your valuation?

The terminal value (TV) =

$$TV_2 = \frac{FCF_3}{r - g} = \frac{150(1.10)}{0.15 - 0.10} = 3,300 \text{ million}$$

- b. Using the given WACC, what is the total firm value of Timex, Inc.?

$$FirmValue = cash_0 + \frac{FCF_1}{1+WACC} + \frac{FCF_2}{(1+WACC)^2} + \frac{TV_2}{(1+WACC)^2}$$

$$FirmValue = 0 + \frac{100}{1.15} + \frac{150}{(1.15)^2} + \frac{3,300}{(1.15)^2} = \$2,695.65 \text{ million}$$

I did not include any cash because it wasn't "excess"

- c. Using your answer from part (b), how much should Swatch, Inc. be willing to pay per share for Timex, Inc.?

Equity Value = firm value–debt = \$2,696 – 900 = \$1,796 million
So, Swatch should be willing to pay \$1,796/100 = \$17.96/share.

- d. If Timex's stock is currently trading at a price of \$19, will Swatch be willing to complete this deal? Why or why not?

No; this deal will not happen.

Swatch currently values Timex at under \$18 per share. It will not be willing to offer anything above that, and Timex shareholders would never sell for less than the current market price, \$19.

- e. Suppose Swatch is worried that its growth assumption is too conservative, and it decides to test the sensitivity of its valuation to using a multiple to calculate the terminal value. To do this, it sees that other watch-making companies, including itself, currently have market value of assets that are about 9 times their EBITDA. If the projected EBITDA for Timex in year 2 is \$400 million, how much will Swatch be willing to pay per share for Timex under this new assumption?

Multiple says $TV_2 = 400 \text{ million} \times 9 = \$3,600 \text{ million}$
Now, resolve for valuation...

$$\text{Firm value} = 0 + \frac{100}{1.15} + \frac{150}{1.15^2} + \frac{3,600}{1.15^2} = \$2,922.50 \text{ million}$$

Subtract \$900 million debt to get equity value = \$2,022.50
Divide by 100 million shares to get price = \$20.23

- f. Would you recommend Swatch try to make a deal?

No right answer! Depends on how confident Swatch is in its estimated FCFs, WACC, and new terminal value.

2. Suppose you've been tasked with valuing Delta Airlines. You've decided, however, that you don't have the time or patience to do a full-blown DCF valuation, and instead want to value Delta using a multiples valuation approach. The two obvious comparable firms are United Airlines (UA) and American Airlines (AA), and for scaling factors, you've decided to try using both total sales and EBITDA. With this goal in mind, you have gathered the following information on UA and AA:

United Airlines

D+E = \$1 billion

Sales = \$1.5 billion

EBITDA = \$0.5 billion

American Airlines

D+E = \$2 billion

Sales = \$1.25 billion

EBITDA = \$0.75 billion

The EBITDA and sales values above are projected values for the current year. Delta, the company you are trying to value, has projected sales of \$2 billion for this year and a projected EBITDA of \$0.75 billion.

- a. What is the average multiple you find using these two comparable companies when using sales as your scaling factor?

For United, its multiple = $1/1.5 = 0.67$

For American, its multiple = $2/1.25 = 1.6$

Average sales multiple = 1.13

- b. What is the average multiple you find using these two comparable companies when using EBITDA as your scaling factor?

For United, its multiple = $1/0.5 = 2$

For American, its multiple = $2/0.75 = 2.67$

Average EBITDA multiple = 2.33

- c. **What is your valuation of Delta when using the sales multiple from part (a), and what is your valuation of Delta when using the EBITDA multiple from part (b)?**

Using average sales multiple of 1.13, the valuation of Delta equals $1.13 \times 2 = \$2.26$ billion.

Using the average EBITDA multiple of 2.33, the valuation of Delta equals $2.33 \times 0.75 = \$1.75$ billion

- d. **Given your answers to part (c), what do you think might be a reasonable valuation for Delta?**

Something between \$1.75 and \$2.26 billion seems reasonable, but it is hard to say much more than that. It's not clear why there is a half billion difference between the two scaling factors, and there is little theory to give us guidance here on which is better here. This is a weakness of multiples.

- e. **Do you think it would be helpful to also add Southwest Airlines and Jet Blue as two additional comparable companies when calculating the average multiple in parts (a) and (b)? Why or why not?**

It's not clear. It's possible that adding these two additional comparables to come up with an average multiple might reduce the noise of our estimates. But, at the same time, it's not clear that these two additional airlines are as comparable since they don't operate international routes and don't have partner alliances. Therefore, adding them may be harmful.